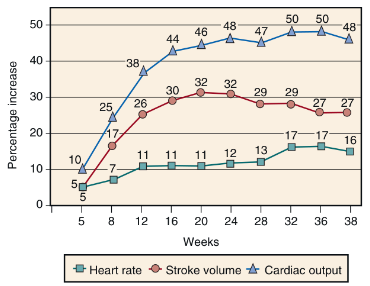

Fig. 58.1 Hemodynamic changes in pregnancy. Shown are the percentage changes from prepregnancy values in heart rate, stroke volume, and cardiac output measured throughout pregnancy. (Modified from Robson SC, Hunter S, Boys RJ, et al. Serial study of factors influencing changes in cardiac output during human pregnancy. *Am J Physiol.* 1989;256:H1060–H1065.)

Fig_58_1

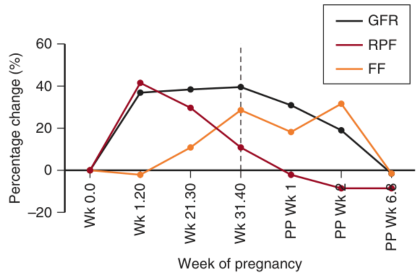

Fig. 58.2 Changes in glomerular filtration *(GFR)*, renal plasma flow *(RPF)*, and filtration fraction *(FF)* during gestation by week *(wk)*. (From Odutayo A, Hladunewich M. Obstetric nephrology: renal hemodynamic and metabolic physiology in normal pregnancy. *Clin J Am Soc Nephrol*. 2012;7:2073–2080[2])

Fig_58_2

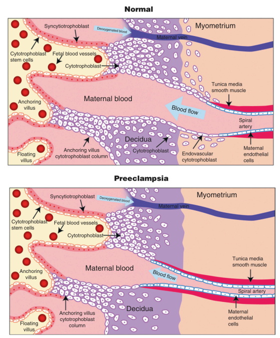

Fig. 58.3 Placentation in normal and preeclamptic pregnancies. In normal placental development *(upper panel),* invasive cytotrophoblasts of fetal origin invade the maternal spiral arteries, transforming them from small-caliber resistance vessels to high-caliber capacitance vessels capable of providing placental perfusion adequate to sustain the growing fetus. During the process of vascular invasion, the cytotrophoblasts differentiate from an epithelial phenotype to an endothelial phenotype, a process referred to as "pseudovasculogenesis" or "vascular mimicry." In preeclampsia *(lower panel),* cytotrophoblasts fail to adopt an invasive endothelial phenotype. Instead, invasion of the spiral arteries is shallow, and they remain small-caliber resistance vessels. (From Lam C, Kim KH, Karumanchi SA. Circulating angiogenic factors in the pathogenesis and prediction of preeclampsia. *Hypertension* 2005;46:1077–1085.)

Fig_58_3

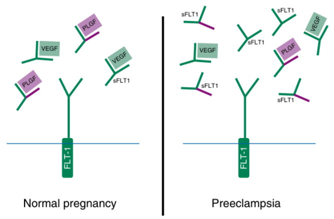

Fig. 58.4 Proposed mechanism of soluble fms-like tyrosine kinase-1 *(sFLT1)*–induced endothelial dysfunction. sFLT1 protein, derived from alternative splicing of FLT1, lacks the transmembrane and cytoplasmic domains but still has the intact vascular endothelial growth factor *(VEGF)* and placental growth factor *(PLGF)* binding extracellular domain. During normal pregnancy, VEGF and PLGF signal through the VEGF receptors *(FLT1)* and maintain endothelial health. In preeclampsia, excess sFLT1 binds to circulating VEGF and PLGF, thus impairing normal signaling of both VEGF and PLGF through their cell surface receptors. Thus excess sFLT1 leads to maternal endothelial dysfunction.

Fig_58_4

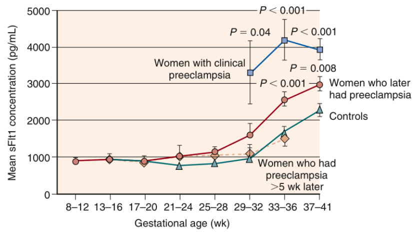

Fig. 58.5 Concentrations of soluble fms-like tyrosine kinase-1 (sFLT1) in preeclampsia and normal pregnancy. Shown are the mean serum sFLT1 concentrations (± standard error of mean) before and after onset of clinical preeclampsia according to the gestational age of the fetus. The *P* values given are for comparisons, after logarithmic transformation, with specimens from controls obtained during the same gestational-age interval. All specimens were obtained before labor and delivery. (From Levine RJ, Maynard SE, Qian C, et al. Circulating angiogenic factors and the risk of preeclampsia. *N Engl J Med.* 2004;350:672–683.)

Fig_58_5

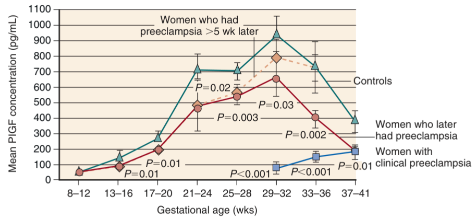

Fig. 58.6 Concentrations of placental growth factor (PlGF) in preeclampsia and normal pregnancy. Shown are the mean serum PlGF concentrations (± standard error of the mean) before and after onset of clinical preeclampsia according to the gestational age of the fetus. The *P* values given are for comparisons, after logarithmic transformation, with specimens from controls obtained during the same gestational-age interval. All specimens were obtained before labor and delivery. (From Levine RJ, Maynard SE, Qian C, et al. Circulating angiogenic factors and the risk of preeclampsia. *N Engl J Med.* 2004;350:672–683.)

Fig_58_6

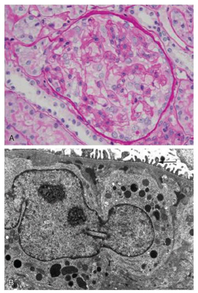

Fig. 58.7 Glomerular endotheliosis. (A) Human preeclamptic glomerulus on light microscopy (periodic acid–Schiff stain). Renal biopsy findings from a 29-year-old woman with twin gestation and severe preeclampsia are shown. The patient's blood pressure was 170/112 mm Hg, and random urine protein-to-creatinine ratio was 9.8. Note the "bloodless" appearance of the glomeruli and absence of the capillary lumen. (Original magnification ×40.) (B) Electron microscopy of biopsy specimen of the glomerulus from the same patient. Note occlusion of capillary lumen cytoplasm and expansion of the subendothelial space with some electron-dense material. Podocyte cytoplasm shows protein resorption droplets and relatively intact foot processes. (Original magnification ×1500.) (Courtesy I. E. Stillman.)

Fig_58_7

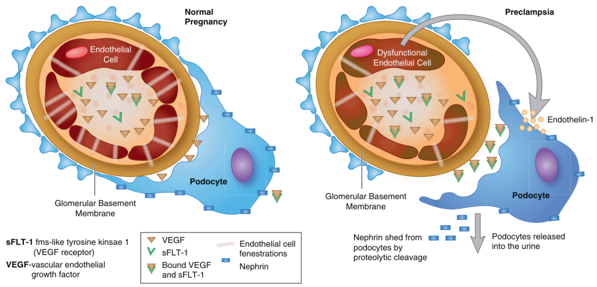

Fig. 58.8 Relationship between podocyte dysfunction and glomerular endothelial cell distribution in preeclampsia. Schema summarizing changes in the glomerulus barrier in normal and preeclamptic pregnancy. (A) In a normal pregnancy, there is an excess of vascular endothelial growth factor VEGF (V shape), compared with the circulating fms-like tyrosine kinase-1 (sFLT-1; *black triangle*) in women who will develop preeclampsia. (B) Placentally derived factors cause endothelial dysfunction, which does not have a direct effect on the podocytes. The circulating SF LT1 does bind to and inhibit the effect of the podocyte-produced VEGF thought to have a major role in maintaining a healthy glomerular barrier. Dysfunctional endothelial cells produce endothelin-1 (ET, *red circles*), which in turn affects the podocytes and induces nephrin shedding *(red rectangle).* The placentally derived factors cause a reduction in endothelial cell fenestrations *(straight lines).* (Adapted from Hennessy A, Makris A. Preeclamptic nephropathy. *Nephrology (Carlton).* 2011;16[2]:134–143.)[88]

Fig_58_8

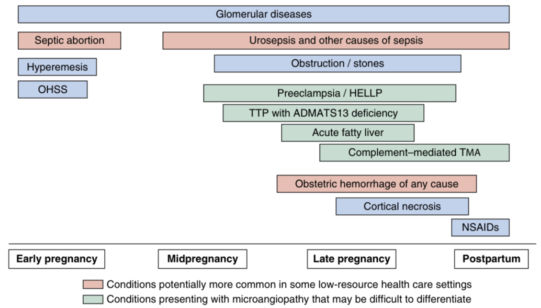

Fig. 58.9 Causes of pregnancy-associated acute kidney injury according to health setting, disease type, and gestational age. *OHSS,* Ovarian hyperstimulation syndrome.

Fig_58_9

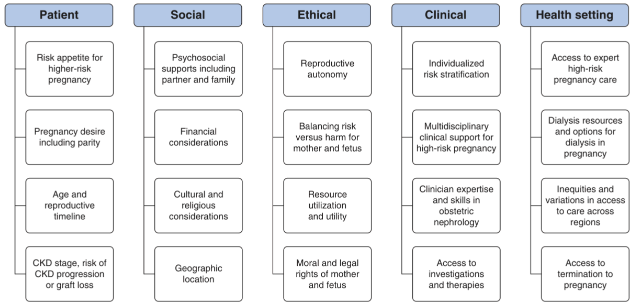

Fig 58.10 Navigating pregnancy with kidney disease: context and broad considerations that require an individualized approach tailored to each woman.

Fig_58_10

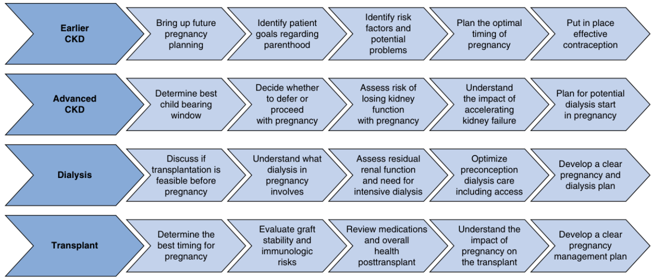

Fig. 58.11 A summary of key decision points and elements for pregnancy planning and care through the spectrum of kidney disease. (From Jesudason S, Williamson A, Huuskes B, Hewawasam E. Parenthood with kidney failure: answering questions patients ask about pregnancy. *Kidney Int Rep*. 2022;7:1477–1492.)

Fig_58_11

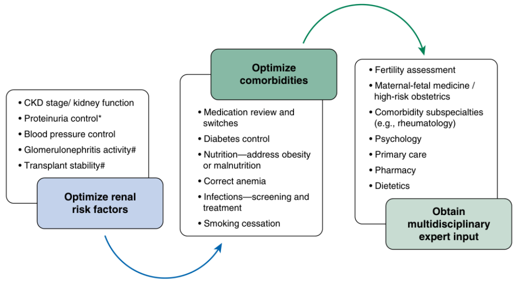

Fig. 58.12 Framework for pregnancy preparation in women with kidney disease considers continuation of renin-angiotensin-aldosterone blockade until conception in selected women where benefit outweighs risk. #Consider biopsy if uncertain.

Fig_58_12

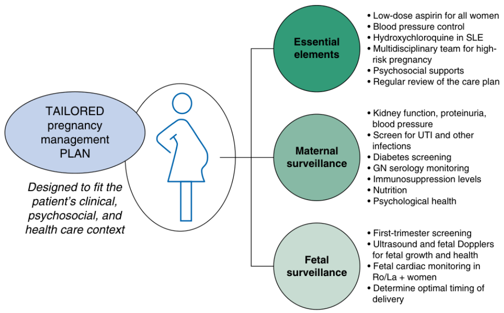

Fig. 58.13 A framework for antenatal care in women with kidney disease. This care plan should be determined preconception and revised at conception and regularly as the pregnancy progresses. The frequency of monitoring and visits will be individualized according to chronic kidney disease stage and pregnancy risk. *SLE*, Systemic lupus erythematosus; *UTI*, urinary tract infection.

Fig_58_13

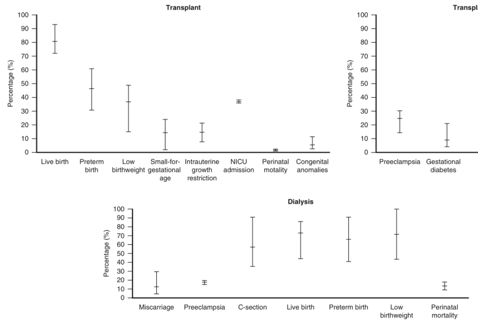

Fig. 58.14 A summary of the range of key maternal and fetal outcomes reported for pregnancies in women with a kidney transplant or receiving dialysis, based on studies large cohort studies drawn from post-2000 contemporaneous cohorts in developed countries.[413,419,420,429,431,442,447-455] Small-for-gestational age is defined as birth weight <10th percentile for gestational age. *NICU,* Neonatal intensive care unit. (Adapted from Jesudason S, Williamson A, Huuskes B, Hewawasam E. Parenthood with kidney failure: answering questions patients ask about pregnancy. *Kidney Int Rep*. 2022;7:1477–1492.)

Fig_58_14

Table 58.1 Physiologic Changes in Pregnancy

Physiologic Variable	Change in Pregnancy
Hemodynamic Parameters	
Plasma volume	Increases 30%-50% above baseline due to RAAS activation, progesterone competitively inhibiting aldosterone, reduced set point for ADH release, and sodium retention of up to 1000 mmol total body sodium
Blood pressure	Decreases by approximately 10 mm Hg below prepregnancy level, with nadir in second trimester around 20 weeks Gradually increases toward prepregnancy levels from 35 weeks
Cardiac output	Increases 30%-50%
Heart rate	Increases by 15-20 beats/min
Renal blood flow	Increases to 50%-80% above baseline; afferent and efferent arteriole vasodilation Increased sensitivity to RAAS
Glomerular filtration rate	150-200 mL/min (increases to 40%-50% above baseline)
Structural changes	Physiologic collecting system dilation and mild hydronephrosis
Serum Chemistry and Hematologic Changes	
Hemoglobin	Decreases by an average of 2 g/L (from 13 to 11 g/L) owing to plasma volume expansion despite increase in red blood cell mass due to increase in erythropoietin
Serum creatinine	Decreases to a nadir of 0.4-0.5 mg/dL in the second trimester and rises back to nonpregnant levels by the third trimester; levels of 0.81-0.87 mg/dL in the second to third trimester represent kidney impairment
Uric acid	Decreases to a nadir of 2.0-3.0 mg/dL by 22-24 weeks due to reduced reabsorption, then increases back to nonpregnant levels toward term
pH	Increases slightly to 7.4-7.45
Partial pressure of carbon dioxide (pCO_2)	Decreases by approximately 10 mm Hg to an average of 27-32 mm Hg
Calcium	Increased calcitriol stimulates increases in both intestinal calcium reabsorption and increased urinary calcium excretion
Serum sodium	Decreases by 4-5 mEq/L below nonpregnancy levels despite increase in total body sodium
Serum osmolality	Decreases to a new osmotic set point of approximately 270 mOsm/kg with reduced ADH (vasopressin) release
Urinary protein	Physiologic proteinuria up to urine protein-to-creatinine ratio of 30 mg/mmol

Table_58_1

Table 58.2 Definitions of Hypertensive Disorder of Pregnancy Categories and Other Types of Hypertension that May be Identified During Pregnancy

HDP Category	Definition

Preeclampsia[a]	Multisystem disorder. New-onset hypertension (≥140 and/or ≥90 mm Hg) after 20 weeks' gestation accompanied by ≥1 of the following new-onset organ involvement items[a]:
	• Kidney involvement
	  • Proteinuria (spot urine protein-to-creatinine ratio ≥30 mg/mmol [0.3 mg/mg] or >300 mg/day in 24-hr urine collection or albumin-to-creatinine ratio of >8 mg/mmol [71 mg/g])
	  • Serum creatinine >90 μmol/L OR 1 mg/dL in the absence of other kidney disease
	• Liver involvement
	  • Raised serum transaminases (in the absence of an alternate diagnosis) with or without right upper quadrant or epigastric pain
	• Hematologic involvement
	  • Thrombocytopenia <150,000/μL
	  • Features of hemolysis: decreased haptoglobin with or without fragmented red cells, elevated lactate dehydrogenase
	  • Disseminated intravascular coagulation
	• Neurologic involvement
	  • Seizure (eclampsia)
	  • Features of cerebral irritability: hyperreflexia with sustained clonus, persistent headache, persistent visual disturbances (photopsia, scotomata, cortical blindness, posterior reversible encephalopathy, retinal vasospasm)
	  • Cerebrovascular accident
	• Pulmonary edema
	• Features of placental dysfunction
	  • Features of fetal growth restriction or deceleration in growth trajectory associated with abnormal umbilical artery Dopplers or oligohydramnios

Gestational hypertension[a]	New onset of hypertension after 20 weeks with no evidence of end-organ involvement
Superimposed preeclampsia*	Features of preeclampsia superimposed on either preexisting hypertension or kidney disease (e.g., sudden increase in blood pressure that was well controlled or escalation of antihypertensive medications to control blood pressure, new-onset proteinuria in a woman with chronic hypertension)
Chronic hypertension	Hypertension confirmed either before pregnancy or before 20 completed weeks' gestation. Chronic hypertension may be either: • Primary (or essential) or • Secondary
White coat hypertension	Where hypertension is present in the office or clinic setting but normal blood pressure readings occur at home or with ambulatory blood pressure monitoring
Masked hypertension	Where there is normal blood pressure in the office or clinic setting but hypertension is noted on home readings or with ambulatory blood pressure monitoring

[a]The signs evident of end-organ dysfunction that developed during pregnancy should return to normal within 3 months of birth.

Table_58_2

Table 58.3 Risk Factors for Preeclampsia

Risk Factor	Pooled Relative Risk (95% CI)
Antiphospholipid antibody syndrome	2.8 (1.8-4.3)[a]
Chronic kidney disease	1.8 (1.5-2.1)[a]
Prior preeclampsia	8.4 (7.1-9.9)[a]
Nulliparity	2.1 (1.9-2.4)[a]
Chronic hypertension	5.1 (4.0-6.5)[a]
Pregestational diabetes mellitus	3.7 (3.1-4.3)[a]
Multiple gestations	2.9 (2.6-3.1)[a]
Strong family history of cardiovascular disease (heart disease/stroke in two or more first-degree relatives)	3.2 (1.4-7.7)[b]
Systemic lupus erythematosus	2.5 (1.0-6.3)[a]
Obesity (prepregnancy BMI >30 kg/m^2)	2.8 (2.6-3.1)[a]
Family history of preeclampsia	2.4 (1.8-3.6)[c]
Advanced maternal age (>40 years)	1.5 (1.2-2.0)[a]
Excessive gestational weight gain (>35 lb)	1.9 (1.7-2.0)[d]
Assisted reproductive technology	1.8 (1.6-2.1)[a]
Previous episode of acute kidney injury	5.9 (3.6-9.7)[e]

[a]From Bartsch E, Medcalf KE, Park AL, et al. High Risk of Pre-eclampsia identification G. Clinical risk factors for pre-eclampsia determined in early pregnancy: systematic review and meta-analysis of large cohort studies. *BMJ*. 2016;353:i1753.
[b]From Ness RB, Markovic N, Bass D, et al. Family history of hypertension, heart disease, and stroke among women who develop hypertension in pregnancy. *Obstet Gynecol.* 2003;102:1366–1371.
[c]From Carr DB, Epplein M, Johnson CO, et al. A sister's risk: family history as a predictor of preeclampsia. *Am J Obstet Gynecol*. 2005;193: 965–972.
[d]From Macdonald-Wallis C, Tilling K, Fraser A, et al. Gestational weight gain as a risk factor for hypertensive disorders of pregnancy. *Am J Obstet Gynecol.* 2013;209:327 e1–e17.
[e]From Tangren JS, Powe CE, Ankers E, et al. Pregnancy outcomes after clinical recovery from AKI. *J Am Soc Nephrol*. 2017;28:1566–1574.

Table_58_3

Table 58.4 Comparison of Clinical and Laboratory Characteristics of the Typical Clinical Syndromes for PE/HELLP, HUS/TTP, and AFLP

	HUS/TTP	PE/HELLP	AFLP
Clinical Characteristic:			
Hemolytic anemia	+++	++	±
Thrombocytopenia	+++	++	− or +
Coagulopathy	−	− or +	+
Neurologic symptoms	++	− or +	− or +
Acute kidney injury severity	+++	+	++
Hypertension	− or +	+++	− or +
Proteinuria	− or +	++	− or +
Elevated AST	− or +	++	+++
Elevated bilirubin	++	+	+++
Anemia	++	+	− or +
Elevated ammonia	−	−	+++

AFLP, Acute fatty liver of pregnancy; *AST,* aspartate aminotransferase; *HELLP,* hemolytic anemia, elevated liver enzymes, and low platelets; *HUS/TTP,* hemolytic uremic syndrome/thrombotic thrombocytopenic purpura; *PE,* Preeclampsia; −, absent; +, mild; ++, moderate; +++, severe or a prominent feature.

Table_58_4

58. Pregnancy and Kidney Disease · assets 19-24 / 24

Table 58.5 Oral Antihypertensives That Can be Used During Pregnancy

Antihypertensives	Class of Agent	Dose (Start From Low Dose and Titrate as Required)	Caution	Side Effects
Oral methyldopa	α Blocker	250-750 mg 3-4 times a day	Avoid in women with a history of depression, anxiety, or postpartum depression	Dry mouth, sedation, abnormal liver function tests, hemolytic anemia
Oral clonidine	α Blocker	75-300 micrograms 3-4 times a day	Risk of rebound hypertension with sudden withdrawal	Dry mouth, sedation
Oral labetalol	α And β blocker	100-400 mg 3-4 times a day	Avoid in women with a history of asthma or chronic airway limitation	Theoretic benefit to uteroplacental perfusion compared with other β blockers
Oral oxprenolol	β Blocker	40-120 mg 3 times a day	Avoid in women with a history of asthma or chronic airway limitation	Bradycardia, fatigue
Oral nifedipine SR	Calcium channel blocker	20-60 mg (slow release) twice a day	Avoid in women with aortic stenosis, may cause peripheral edema	Peripheral edema, headaches, palpitations
Oral nifedipine IR	Calcium channel blocker	10-30 mg (immediate release) 3 times a day	Avoid in women with aortic stenosis, may cause peripheral edema	Peripheral edema, headaches, palpitations
Oral hydralazine	Vasodilator	12.5-50 mg 3-4 times a day	May cause tachycardia if given as first line (without concurrent α, β, or calcium blockade)	Tachycardia, headache, lupus-like syndrome, sweating, shortness of breath

IR, Immediate release; *SR,* sustained release.

Table_58_5

Table 58.6 Agents for Use for Severe Hypertension in Pregnancy

Antihypertensives	Class of Agent	Onset of Action	Dose (Start From Low Dose and Titrate as Required)	Side Effects or Considerations
Oral nifedipine (IR)	Calcium channel blocker	30-45 minutes	10-20 mg every 30 minutes maximum of 45 mg	Headache, tachycardia, peripheral edema with recurrent usage
IV hydralazine	Vasodilator	15-20 minutes	5-10 mg every 20 minutes maximum of 30 mg	Tachycardia, headache
IV labetalol	β blocker	5 minutes	20-40 mg every 10-15 minutes, maximum of 80 mg	Bradycardia, shortness of breath Avoid in women with asthma or chronic airway disease
IV diazoxide	Vasodilator	3-5 minutes	15 mg every 5-10 minutes	Tachycardia, palpitations
Oral methyldopa[a]	α Blocker	30-120 minutes	1000 mg as a single dose	Dry mouth, sedation, abnormal liver function tests. Avoid repeated use in women with mental health issues—associated with postnatal depression
Oral labetalol[a]	β Blocker	30-120 minutes	200 mg up to 4 doses every hour	Bradycardia, shortness of breath Avoid in women with asthma or chronic airway disease

[a]Trialed in low-resource setting. Consider where intravenous access cannot be established or is not possible.

Table_58_6

Table 58.7 Best-Practice Approach to Pregnancy Counseling in Women With Kidney Disease

Domain	Suggested Approach
Timing	Raise potential motherhood as early as feasible to allow planning Prospectively discuss the best window for pregnancy Allow sufficient time for evolution of discussions over multiple visits Review and revisit discussions at regular intervals
Communication	Support the woman's right to pursue pregnancy (or not) Avoid making women defend their choices Avoid judgmental comments or "forbidding" pregnancy Explain risks without catastrophizing Provide hope where possible Identify and include any other key persons (partner, family) Provide reassurance that care will be given
Patient values	Identify and acknowledge patient goals Do not assume motherhood is desired by all Acknowledge grief related to limitations to motherhood Understand how fears are balanced with desire for parenthood Define external pressures, obligations, and feelings of guilt
Decision making	Acknowledge the decisional burden Identify how much decisional control women want Assess risk based on individual clinical context Understand how risks and decisions are rationalized Determine individual appetite for "risk" Facilitate autonomy and decisional ownership Adopt shared decision making approaches
Information	Identify how much information women want to have Discuss maternal and fetal risks, long-term health impact, potential pregnancy outcomes, likely pregnancy management and progress Refer to other services (obstetric, maternal-fetal medicine, genetic, reproductive medicine) for additional information and counseling Actively facilitate and address questions

From Jesudason S, Williamson A, Huuskes B, Hewawasam E. Parenthood with kidney failure: answering questions patients ask about pregnancy. *Kidney Int Rep*. 2022;7:1477–1492.

Table_58_7

Table 58.8 Immunosuppressive Medications in Pregnancy

Drug	Recommendations
Prednisone	Safe when used long term at low to moderate doses (5-10 mg/day) Safe when given acutely at high doses for treatment of acute rejection or induction therapy for glomerular disease
Cyclosporine, tacrolimus	Extensive clinical data suggest safe at low-to-moderate clinical doses Changes in absorption and metabolism require close monitoring of levels and frequent dose adjustments in pregnancy; however, therapeutic range may need to be lower to avoid toxicity from active drugs. The active (unbound) fraction rises in response to physiologic changes in hematocrit and albumin in pregnancy. Whole blood levels need careful interpretation.
Azathioprine	Considered safe at dosages <2 mg/kg per day, but at higher doses associated with fetal growth restriction
Mycophenolate mofetil	Contraindicated—teratogenic in animal and human studies, associated with early pregnancy loss and fetal defects
Sirolimus, everolimus	Contraindicated—embryo/fetal toxicity in rodents; human studies are lacking beyond case reports/anecdotal data
Rituximab (anti-CD20 mAb)	Use in pregnancy is growing, no clear signal of adverse fetal effects. May be used before conception or during pregnancy where benefit outweighs potential risks
Muromonab-CD3 (OKT-3)	Case reports of successful use for induction in unsuspected pregnancy and for acute rejection, but data are limited
Antithymocyte globulin (C)	Avoid in pregnancy; Very limited animal and human data
Belatacept, basiliximab, alemtuzumab	Avoid in pregnancy; inadequate safety data in humans

Drawn from selected references.[336,347,368,380,381,409,437,456-472]

Table_58_8

Table 58.9 Recommended Medical Management for Pregnant Women With Kidney Failure

Dialysis Recipient	Transplant Recipient
Model of Care	
• Establish a team including nephrologists, obstetricians with maternal-fetal medicine and high-risk pregnancy experience, midwives and allied health support including dietetics, psychology, and pharmacy. Identify the coordinator of care. • Care close to a tertiary center for expert nephrology, transplant, and obstetric care	
Medications	
• Stop phosphate binders, calcimimetics • Switch to pregnancy-safe antihypertensives • Commence low-dose aspirin 75-150 mg daily	• Switch to pregnancy-safe antihypertensives and immunosuppression • Cease antiviral therapy • Commence low-dose aspirin 75-150 mg daily
Modality-Specific Management	
• Measure residual function • Consider PD switch to HD where feasible • Titrate dialysis hours and frequency based on residual function; aim for predialysis urea 12 mmol/L • Implement intensive dialysis regimen (>36 hours/week, titrated to predialysis urea) in women with no residual function • Do not use Kt/V to assess clearances • Use minimal anticoagulation • Regularly review and titrate dialysate composition based on bicarbonate, potassium, calcium, and phosphate blood levels • Adjust peritoneal dialysis prescription to increase clearances; reduce volume if required for patient comfort • Frequently review and adjust dry weight by 0.3-0.5 kg/week from second trimester • Monitor dialysis access closely for infection and complications • Minimum weekly clinical review	• Minimum monthly clinical review and measurement of kidney function, proteinuria, and immunosuppressive trough levels • Increase frequency of review as gestation progresses • Anticipate a fall in whole blood (bound) trough CNI levels in the second trimester. Interpret falling CNI levels cautiously and beware of potential toxicity after dose increases • Consider transplant biopsy if acute graft dysfunction without clinical suspicion of preeclampsia; balance value of biopsy against gestational age and likelihood of early delivery
Blood Pressure	
• Recommend home BP monitor • Vigilance for superimposed preeclampsia • Consider sFLT1/PIGF levels where available • Aim for BP <135/85 • Avoid placental hypoperfusion from intradialytic hypotension	• Recommend home BP monitor • Vigilance for superimposed preeclampsia • Value of sFLT1/PIGF levels uncertain • Aim for BP <135/85 • Avoid hypotension <110/70
Anemia	
• Use ESA to achieve target Hb 110 g/dL • Ensure iron stored is replete according to local protocols • Develop criteria for blood transfusion—optimize hemoglobin to avoid transfusion	
Infection	
• Monitor closely for catheter sepsis or AVF infection • Screen for and treat bacteriuria/UTI; consider prophylaxis • Coronavirus vaccination	• Screen for and treat bacteriuria/UTI; consider prophylaxis • Coronavirus vaccination • Screen/Monitor for CMV in women at risk (past CMV, previous prophylaxis)
Nutrition	
• Start or increase dose of vitamins B and C and folic acid supplements • Reduce fluid, potassium and phosphate restrictions; may require supplementation • Expert dietitian involvement to optimize protein and calorie intake	• Dietitian review as required
Diabetes Screening	
• Early screening (16-20 weeks) in high-risk patients (prednisolone, tacrolimus, family history, past gestational diabetes mellitus, obesity) • Routine screening at 28 weeks' gestation or per local practice	
Diabetes Management	
• Diabetes education and dietitian • Insulin therapy	• Diabetes education and dietitian • Metformin or insulin

Continued on following page

Table_58_9_part1

Table 58.9 Recommended Medical Management for Pregnant Women With Kidney Failure (Continued)

Dialysis Recipient	Transplant Recipient
Fetal Monitoring	
• Dating, morphology scans, and first-trimester screening as per local practice • Consider noninvasive prenatal testing (cell-free DNA) for aneuploidy • Frequent (fortnightly) growth scans in third trimester to monitor growth restriction, amniotic fluid index, and vascular changes suggesting preeclampsia	
Delivery and Early Postpartum Care	
• Stop aspirin at 35 weeks' gestation • Develop and regularly review the "goalposts" for delivery including timing • Trial of labor is not contraindicated • Clarify transplant anatomy before delivery • Steroid "stress dose" if on chronic glucocorticoid therapy • Avoid nonsteroidal antiinflammatory medications for pain relief • Consider venous thromboembolism prophylaxis following operative delivery • Urgent postdelivery transplant imaging for any reduction in function or urine output • Establish breastfeeding if desired by mother; support decisions about breastfeeding	
Postpregnancy Care	
• Contraception in place • Psychosocial support • Resume nonpregnant dialysis prescription • Monitor residual kidney function	• Contraception in place • Psychosocial support • Close monitoring of graft function and CNI levels • Adjust/reinstate medications

AVF, Arteriovenous fistula; *BP,* blood pressure; *CMV,* cytomegalovirus; *CNI,* calcineurin inhibitor; *PIGF,* placental growth factor; *sFLT1,* soluble fms-like tyrosine kinase 1; *UTI,* urinary tract infection.
Reproduced from Jesudason S, Williamson A, Huuskes B, Hewawasam E. Parenthood with kidney failure: answering questions patients ask about pregnancy. *Kidney Int Rep.* 2022;7:1477–1492.

Table_58_9_part2